

Extended Relational Algebra: GROUP BY, HAVING, ORDER BY

With Aggregate Functions and Examples

Classical relational algebra does not support grouping, aggregation, or ordering. In **extended relational algebra**, additional operators are introduced:

γ (grouping + aggregation), σ (selection), τ (sorting)

1. GROUP BY (γ) Operator

The γ operator performs grouping and aggregation.

$$\gamma_{\text{group_attrs, agg_funcs}}(R)$$

Example:

SQL:

```
SELECT dept_name, COUNT(*) AS num_instructors
FROM instructor
GROUP BY dept_name;
```

Relational Algebra:

$$\gamma_{\text{dept_name, COUNT(*)} \rightarrow \text{num_instructors}}(\text{instructor})$$

2. HAVING Clause (Selection after Grouping)

The HAVING condition is applied *after* grouping and aggregation, using σ .

Example:

SQL:

```
SELECT dept_name, COUNT(*) AS num_instructors
FROM instructor
GROUP BY dept_name
HAVING COUNT(*) > 5;
```

Relational Algebra:

$$\sigma_{\text{num_instructors} > 5}(\gamma_{\text{dept_name, COUNT(*)} \rightarrow \text{num_instructors}}(\text{instructor}))$$

3. ORDER BY (τ) Operator

The τ operator sorts tuples by one or more attributes.

$$\tau_{\text{sort_attr} [\text{ASC} \text{---} \text{DESC}]}(R)$$

Example:

SQL:

```
SELECT name, salary
FROM instructor
ORDER BY salary DESC;
```

Relational Algebra:

$$\tau_{\text{salary DESC}}(\pi_{\text{name,salary}}(\text{instructor}))$$

4. Combined Example

SQL Query

```
SELECT dept_name, COUNT(*) AS num_instructors
FROM instructor
GROUP BY dept_name
HAVING COUNT(*) > 5
ORDER BY num_instructors DESC;
```

Relational Algebra:

$$\tau_{\text{num_instructors DESC}}(\sigma_{\text{num_instructors} > 5}(\gamma_{\text{dept_name, COUNT(*)} \rightarrow \text{num_instructors}}(\text{instructor})))$$

5. SQL $\rightarrow$ Relational Algebra Mapping

SQL Clause	Relational Algebra Equivalent
GROUP BY	$\gamma_{\text{group_attrs, agg_funcs}}(R)$
HAVING	$\sigma_{\text{condition}}(\gamma_{\dots}(R))$
ORDER BY	$\tau_{\text{attr} [\text{ASC} \text{---} \text{DESC}]}(R)$

6. Execution Flow Diagram

